

Adam Optimizer

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Chapter 1

General Setup and Technology Tools

1.1 Adam

Definition 1.1.1 (Innovation).

Let $\mathcal{U}$ be a measurable space. A pair (X, U) consisting of

- (i) a $\mathcal{U}$ -valued random variable U and
- (ii) a product measurable mapping $X: \mathcal{U} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$

is called *innovation*.

$\mathcal{U}$ is the random samples. And X puts $\mathcal{U}$ into stochastic gradient.

Definition 1.1.2 (Adam optimizer).

Let

- (i) $\alpha, \beta \in [0, 1)$, $\epsilon \in (0, \infty)$ with $\alpha < \sqrt{\beta}$ (the damping parameters),
- (ii) $n_0 \in \mathbb{N}_0$, $\theta_{n_0}, m_{n_0} \in \mathbb{R}^d$, $v_{n_0} \in [0, \infty)^d$ (the initialisation),
- (iii) a decreasing $(0, \infty)$ -valued sequence $(\gamma_n)_{n \in \mathbb{N}}$ (the sequence of step-sizes) and
- (iv) (X, U) an innovation.

An $\mathbb{R}^d$ -valued stochastic process $(\theta_n)_{n \in \mathbb{N}_0 \cap [n_0, \infty)}$ is called Adam algorithm with damping parameters $(\alpha, \beta, \epsilon)$ and step-sizes (γ_n) started at time n_0 in $(\theta_{n_0}, m_{n_0}, v_{n_0})$, if it satisfies for every $n \in \mathbb{N} \cap (n_0, \infty)$ and $i = 1, \dots, d$

$$\theta_n^{(i)} = \theta_{n-1}^{(i)} + \gamma_n \sigma_n^{(i)} m_n^{(i)}, \quad (1.1)$$

where $(U_n)_{n \in \mathbb{N} \cap (n_0, \infty)}$ is a family of independent copies of U and for $n \in \mathbb{N} \cap (n_0, \infty)$ and $i = 1, \dots, d$,

- $m_n = \alpha m_{n-1} + (1 - \alpha) X(U_n, \theta_{n-1})$,
- $v_n^{(i)} = \beta v_{n-1}^{(i)} + (1 - \beta) (X^{(i)}(U_n, \theta_{n-1}))^2$ and
- $\sigma_n^{(i)} = \frac{1}{\sqrt{v_n^{(i)} / (1 - \beta^n) + \epsilon}}$.

We call the sequence $(t_n)_{n \in \mathbb{N}_0}$ given by

$$t_n = \sum_{k=1}^n \gamma_k \quad (1.2)$$

the training times of the Adam algorithm.

1.2 Adam sequence space

Consider a sequence of history:

$$\mathbf{x} = (x_k)_{k \in -\mathbb{N}_0} = (x_0, x_{-1}, x_{-2}, \dots)$$

Consider an operator: The whole history sequence $\Rightarrow$ Adam direction

Definition 1.2.1.

Let $\alpha, \beta, \epsilon \in [0, \infty)$ with $\alpha < \sqrt{\beta} < 1$ and let $(\varrho_k)_{k \in -\mathbb{N}_0}$ as in (1.4).

$$g: \ell_{\varrho}^d \rightarrow \mathbb{R}^d, \quad \mathbf{x} = (x_k)_{k \in -\mathbb{N}_0} \mapsto \left(\frac{(1-\alpha) \sum_{k \in -\mathbb{N}_0} \alpha^{-k} x_k^{(i)}}{\epsilon + \sqrt{(1-\beta) \sum_{k \in -\mathbb{N}_0} \beta^{-k} (x_k^{(i)})^2}} \right)_{i \in \{1, \dots, d\}} \quad (1.3)$$

One more natural way is:

$$g(\mathbf{x}) = \frac{m(\mathbf{x})}{\sqrt{v(\mathbf{x})} + \epsilon}$$

We are curious about how *sensitive* g is to the change of x_k in $\mathbf{x}$.

One natural idea is to find the partial derivative of g :

$$\begin{aligned} \partial_{x_k} &= \frac{(1-\alpha)\alpha^{-k}}{\sqrt{v(\mathbf{x})} + \epsilon} - \frac{(1-\beta)\beta^{-k}m(\mathbf{x})x_k}{(\sqrt{v(\mathbf{x})} + \epsilon)^2 \sqrt{v(\mathbf{x})}} \\ &\leq (1-\alpha)\epsilon^{-1}(\alpha^{-k} + \frac{1}{\sqrt{1-\alpha^2/\beta}}\beta^{-\frac{k}{2}}) \end{aligned}$$

So

$$|g(\mathbf{x}) - g(\mathbf{y})| \leq \sum_{k \in -\mathbb{N}_0} (1-\alpha)\epsilon^{-1}(\alpha^{-k} + \frac{1}{\sqrt{1-\alpha^2/\beta}}\beta^{-\frac{k}{2}})x_k$$

Thus, we can naturally define the norm:

Definition 1.2.2 (Adam sequence space).

Let $\alpha, \beta \in [0, 1)$, $\epsilon \in (0, \infty)$ with $\alpha < \sqrt{\beta}$ be damping factors. We let

$$\varrho_k = (1-\alpha)\epsilon^{-1}\left(\alpha^{-k} + \frac{1}{\sqrt{1-\alpha^2/\beta}}\beta^{-k/2}\right), \quad \text{for all } k \in -\mathbb{N}_0, \quad (1.4)$$

and denote by ℓ_{ϱ}^d the space of all sequences $\mathbf{x} = (x_k)_{k \in -\mathbb{N}_0} \in (\mathbb{R}^d)^{-\mathbb{N}_0}$ with

$$\|\mathbf{x}\|_{\ell_{\varrho}^d} := \sum_{k \in -\mathbb{N}_0} \varrho_k |x_k| < \infty. \quad (1.5)$$

We equip the space with the norm $\|\cdot\|_{\ell_\varrho^d}$. Moreover, we let for a tuple $(m, v) \in \mathbb{R}^d \times [0, \infty)^d$

$$(m, v)_{\ell_\varrho^d} = \inf \left\{ \|\mathbf{x}\|_{\ell_\varrho^d} : \mathbf{x} \in \ell_\varrho^d, m = (1 - \alpha) \sum_{k \in -\mathbb{N}_0} \alpha^{-k} x_k \text{ and } v^{(i)} = (1 - \beta) \sum_{k \in -\mathbb{N}_0} \beta^{-k} (x_k^{(i)})^2 \text{ for } i = 1, \dots, d \right\}, \quad (1.6)$$

where the infimum of the empty set is ∞ .

Lemma 1.2.3. *Let $\alpha, \beta, \epsilon \in [0, \infty)$ with $\alpha < \sqrt{\beta} < 1$ and let $(\varrho_k)_{k \in -\mathbb{N}_0}$ and g as defined in (1.4) and (1.3), respectively. Then g is Lipschitz continuous with Lipschitz constant 1 and g is uniformly bounded by $\sqrt{d} \frac{1-\alpha}{\sqrt{1-\beta}} \frac{1}{\sqrt{1-\alpha^2/\beta}}$.*

Lemma 1.2.4. *For every $k \in -\mathbb{N}_0$, one has that $\varrho_{k-1} \leq \sqrt{\beta} \varrho_k$ and the translation operator*

$$T: \ell_\varrho^d \rightarrow \ell_\varrho^d, \quad \mathbf{x} = (x_k)_{k \in -\mathbb{N}_0} \mapsto (\mathbb{1}_{\{k \neq 0\}} x_{k+1})_{k \in -\mathbb{N}_0} \quad (1.7)$$

is Lipschitz continuous with Lipschitz constant $\sqrt{\beta} < 1$. In particular, it maps ℓ_ϱ to ℓ_ϱ .

1.2.1 Proof of Lemma 1.2.3

Proof. We first assume that $d = 1$ and briefly write $\ell_\varrho = \ell_\varrho^1$ and $\|\cdot\|_{\ell_\varrho} = \|\cdot\|_{\ell_\varrho^1}$. For $\mathbf{x} \in \ell_\varrho$, let

$$m(\mathbf{x}) = (1 - \alpha) \sum_{k \in -\mathbb{N}_0} \alpha^{-k} x_k \text{ and } v(\mathbf{x}) = (1 - \beta) \sum_{k \in -\mathbb{N}_0} \beta^{-k} x_k^2 \quad (1.8)$$

so that

$$g(\mathbf{x}) = \frac{m(\mathbf{x})}{\sqrt{v(\mathbf{x})} + \epsilon}. \quad (1.9)$$

Using Cauchy-Schwarz and the fact that $\alpha < \sqrt{\beta}$ we get that for all $\mathbf{x} \in \ell_\varrho$

$$\begin{aligned} |m(\mathbf{x})| &= \left| (1 - \alpha) \sum_{k=-\infty}^0 (\alpha/\sqrt{\beta})^{-k} \sqrt{\beta}^{-k} x_k \right| \leq (1 - \alpha) \left(\sum_{k=0}^{\infty} (\alpha/\sqrt{\beta})^{2k} \right)^{1/2} \left(\sum_{k=-\infty}^0 \beta^{-k} x_k^2 \right)^{1/2} \\ &\leq \frac{1 - \alpha}{\sqrt{1 - \beta}} \frac{1}{\sqrt{1 - \alpha^2/\beta}} \sqrt{v(\mathbf{x})} \end{aligned} \quad (1.10)$$

which entails the uniform bound for g that is stated in the lemma for $d = 1$. The multivariate estimates follows immediately by applying this estimate onto the individual components.

To prove the Lipschitz continuity we again focus at first on the case $d = 1$. Fix $n \in \mathbb{N}$ and let $\mathbf{x} = (x_0, \dots, x_{-n}, 0, \dots) \in \ell_\varrho$. One has for all $k = 0, \dots, -n$ that

$$\partial_{x_k} g(\mathbf{x}) = (1 - \alpha) \alpha^{-k} \frac{1}{\sqrt{v(\mathbf{x})} + \epsilon} - (1 - \beta) \beta^{-k} \frac{m(\mathbf{x})}{(\sqrt{v(\mathbf{x})} + \epsilon)^2} \frac{1}{\sqrt{v(\mathbf{x})}} x_k \quad (1.11)$$

so that by using (1.10) and $\sqrt{v(\mathbf{x})} \geq \sqrt{1 - \beta} \beta^{-k/2} |x_k|$ we get that

$$|\partial_{x_k} g(\mathbf{x})| \leq (1 - \alpha) \epsilon^{-1} \alpha^{-k} + \frac{1 - \alpha}{\sqrt{1 - \alpha^2/\beta}} \epsilon^{-1} \beta^{-k/2} = \varrho_k. \quad (1.12)$$

For another $\tilde{\mathbf{x}} = (\tilde{x}_0, \dots, \tilde{x}_{-n}, 0, \dots) \in \ell_\varrho$ we let, for $\ell = 0, \dots, n + 1$,

$$\mathbf{x}^\ell = (\tilde{x}_0, \dots, \tilde{x}_{-\ell+1}, x_{-\ell}, \dots, x_{-n}, 0, \dots). \quad (1.13)$$

Note that $\mathbf{x}^0 = \mathbf{x}$ and $\mathbf{x}^{n+1} = \tilde{\mathbf{x}}$ and we get with the triangle inequality and the mean value theorem that

$$|g(\mathbf{x}) - g(\tilde{\mathbf{x}})| \leq \sum_{\ell=0}^n |g(\mathbf{x}^\ell) - g(\mathbf{x}^{\ell+1})| \leq \sum_{\ell=0}^n \varrho_{-\ell} |x_{-\ell} - \tilde{x}_{-\ell}| = \|\mathbf{x} - \tilde{\mathbf{x}}\|_{\ell_\varrho}. \quad (1.14)$$

This inequality remains true for general $\mathbf{x}, \tilde{\mathbf{x}} \in \ell_\varrho$ since

$$g(\mathbf{x}) = \lim_{n \rightarrow \infty} g(x_0, \dots, x_{-n}, 0, \dots). \quad (1.15)$$

Indeed, this entails that

$$|g(\mathbf{x}) - g(\tilde{\mathbf{x}})| \leq \limsup_{n \rightarrow \infty} |g(x_0, \dots, x_{-n}, 0, \dots) - g(\tilde{x}_0, \dots, \tilde{x}_{-n}, 0, \dots)| \leq \|\mathbf{x} - \tilde{\mathbf{x}}\|_{\ell_\varrho}. \quad (1.16)$$

To obtain the result in the multivariate setting we proceed as above. Fix $n \in \mathbb{N}$, $\mathbf{x} = (x_0, \dots, x_{-n}, 0, \dots) \in \ell_\varrho^d$ and $\tilde{\mathbf{x}} = (\tilde{x}_0, \dots, \tilde{x}_{-n}, 0, \dots) \in \ell_\varrho^d$ and define for every $\ell = 0, \dots, n + 1$, $\mathbf{x}^\ell \in \ell_\varrho^d$ accordingly. Then one has for every $\ell = 0, \dots, n$ that

$$|g(\mathbf{x}^\ell) - g(\mathbf{x}^{\ell+1})| = \left(\sum_{i=1}^d |g(\mathbf{x}^{\ell,(i)}) - g(\mathbf{x}^{\ell+1,(i)})|^2 \right)^{1/2} \leq \varrho_{-\ell} \left(\sum_{i=1}^d |x_\ell^{(i)} - \tilde{x}_\ell^{(i)}|^2 \right)^{1/2}, \quad (1.17)$$

where we applied (1.12) for each of the d components. This entails the multivariate variant of estimate (1.14) that is

$$|g(\mathbf{x}) - g(\tilde{\mathbf{x}})| \leq \|\mathbf{x} - \tilde{\mathbf{x}}\|_{\ell_\varrho^d}. \quad (1.18)$$

The rest follows in complete analogy to the one dimensional setting. $\square$

1.2.2 Proof of Lemma 1.2.4

Proof. By definition of (ϱ_k) it immediately follows that, for every $k \in -\mathbb{N}_0$,

$$\varrho_{k-1} \leq \sqrt{\beta} \varrho_k. \quad (1.19)$$

Consequently, one has that, for every $\mathbf{x} \in \ell_\varrho^d$,

$$\|T(\mathbf{x})\|_{\ell_\varrho^d} = \sum_{k \in -\mathbb{N}_0} \varrho_{k-1} |x_k| \leq \sqrt{\beta} \sum_{k \in -\mathbb{N}_0} \varrho_k |x_k| = \sqrt{\beta} \|\mathbf{x}\|_{\ell_\varrho^d}. \quad (1.20)$$

The statement follows by linearity of T . $\square$

1.3 Adam vector field

Definition 1.3.1 (Adam vector field).

For a tuple $(\alpha, \beta, \epsilon)$ of damping parameters and an innovation (X, U) we call the mapping $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ given by

$$f^{(i)}(\theta) = (1 - \alpha) \mathbb{E}^\theta \left[1 / \left(\sqrt{(1 - \beta) \sum_{k \in -\mathbb{N}_0} \beta^{-k} X^{(i)}(U_k, \theta)^2} + \epsilon \right) \sum_{k \in -\mathbb{N}_0} \alpha^{-k} X^{(i)}(U_k, \theta) \right] \quad (1.21)$$

with $(U_k)_{k \in -\mathbb{N}_0}$ being a family of independent copies of U , the Adam vector field of the innovation (X, U) for the damping parameters $(\alpha, \beta, \epsilon)$.

Chapter 2

Analysis of Adam evolution and approximation methods

2.1 Analysis of Adam with fixed θ

We study the stochastic optimization algorithm Adam introduced in [40]; its convergence behaviour for general step-size sequences is analysed in [17]. Our analysis follows the classical stochastic approximation framework of [58].

We consider how Adam get update-data(1.1.1 and 1.1.2):

$$X_n = X(U_n, \theta_{n-1})$$

However, it's very hard to give an analysis with such *difficult* setup. We could first fix θ , and see how Adam behaves:

Fix $\theta \in \mathbb{R}^d$ and let $(U_k)_{k \in \mathbb{Z}}$ be an i.i.d. sequence of copies of U . Moreover, for every $n \in \mathbb{Z}$, let $X_n = X(U_n, \theta)$ and $\mathbf{X}(n) = (X_{n+k})_{k \in -\mathbb{N}_0}$. We fix $n_0 \in \mathbb{N}_0$ and consider the process $(\bar{\Theta}_n)_{n \geq n_0}$ given by

$$\bar{\Theta}_n = \sum_{k=n_0+1}^n \gamma_k g(\mathbf{X}(k)). \quad (2.1)$$

Thus, only $\mathbf{X}$ generated in θ and the stochastic term will drive Adam.

Let Adam walk from $n_0 + 1$ to n . Lemma 2.1.1 will show how *far* $\bar{\Theta}_n$ will be away from it's expection. And Lemma 2.1.2 will analysis the max *error* of the whole path.

Lemma 2.1.1. *Let $p \in [2, \infty)$. One has for every $n \in \mathbb{N}_0$ with $n \geq n_0$ that*

$$\mathbb{E}[|\bar{\Theta}_n - \mathbb{E}[\bar{\Theta}_n]|^p]^{2/p} \leq \kappa^2 \sum_{k=n_0+1}^n \gamma_k^2 \mathbb{E}[|X(U, \theta) - \mathbb{E}[X(U, \theta)]|^p]^{2/p}, \quad (2.2)$$

where

$$\kappa = 2C_p((1 - \sqrt{\beta})^{-2} \varrho_0 \|\varrho\|_{\ell_1} + \|\varrho\|_{\ell_1}^2), \quad (2.3)$$

ϱ is as in (1.4) and C_p is the constant appearing in the the Burkholder-Davis-Gundy inequality for the p -th moment.

Lemma 2.1.2. Let $p \in (2, \infty)$. One has for every $n \in \mathbb{N}_0$ with $n \geq n_0$

$$\mathbb{E} \left[\max_{k=n_0, \dots, n} |\bar{\Theta}_k - \mathbb{E}[\bar{\Theta}_k]|^p \right]^{1/p} \leq (1 - 2^{-(\frac{1}{2} - \frac{1}{p})})^{-1} \kappa \gamma_{n_0+1} \sqrt{n - n_0} \mathbb{E}[|X_0 - \mathbb{E}[X_0]|^p]^{1/p}, \quad (2.4)$$

where κ is as in (2.3).

2.1.1 Proof of Lemma 2.1.1

Our goal is to give an analysis of

$$\bar{\Theta}_n - \mathbb{E}[\bar{\Theta}_n] \quad (2.5)$$

and we wish to have some form like:

$$\|\bar{\Theta}_n - \mathbb{E}[\bar{\Theta}_n]\|_{L^p} \lesssim \left(\sum_{k=n_0+1}^n \gamma_k^2 \right)^{1/2} \|X - \mathbb{E}X\|_{L^p}$$

(below we will analysis why $(\sum_{k=n_0+1}^n \gamma_k^2)^{1/2}$ is a natural estimation)

Doob Martingale Decomposition

We would consider the Doob Martingale and a stopping time N :

$$M_n = \mathbb{E}[\bar{\Theta}_N | \mathcal{F}_n] - \mathbb{E}[\bar{\Theta}_N] \quad (2.6)$$

As $\mathcal{F}_n$ increases (or say: $\dots, X_{n-1}, X_n, X_{n+1}, \dots$ get known *one by one*), we could analysis how each term influence the total error estimation:

$$\begin{aligned} \Delta M_n &= M_n - M_{n-1} \\ &= \mathbb{E}[\bar{\Theta}_N | \mathcal{F}_n] - \mathbb{E}[\bar{\Theta}_N | \mathcal{F}_{n-1}] \\ &= \mathbb{E} \left[\sum_{r=n_0+1}^N \gamma_r g(\mathbf{X}(r)) | \mathcal{F}_n \right] - \mathbb{E} \left[\sum_{r=n_0+1}^N \gamma_r g(\mathbf{X}(r)) | \mathcal{F}_{n-1} \right]. \end{aligned}$$

Proof. For ease of notation we prove the statement for $d = 1$. The generalisation to general d is straight-forward. Let $\tilde{X}_0$ be identically distributed as X_0 and independent of the sequence $(X_n)_{n \in \mathbb{Z}}$ and set $\tilde{X}_n = X_n$ for $n \in \mathbb{Z} \setminus \{0\}$. We let

$$\tilde{\mathbf{X}}(n) = (\tilde{X}_{n+k})_{k \in -\mathbb{N}_0}. \quad (2.7)$$

Thus $\tilde{\mathbf{X}}(n)$ agrees with $\mathbf{X}(n)$ in all but at most one component.

Why we need to change $\mathbf{X}(n)$ to $\tilde{\mathbf{X}}(n)$?

To estimate ΔM_n , we are just asking

How big the influence of the n -th random variable X_n is to the final result

So $\tilde{\mathbf{X}}(n)$ agrees with $\mathbf{X}(n)$ in all but at most one component.

This is the thought of standrand replacement (or *Efron-Stein*):

The importance of a random variable

$\Downarrow$

change it to an independent copy, how the output change

We let $(\mathcal{F}_n)_{n \in \mathbb{Z}}$ be the filtration generated by (X_n) and consider for fixed $N \geq n_0$ the martingale $(M_n)_{n \in \mathbb{Z}}$ given by

$$M_n = \mathbb{E}[\bar{\Theta}_N | \mathcal{F}_n] - \mathbb{E}[\bar{\Theta}_N] \quad (2.8)$$

We denote by $([M]_n)_{n \in \mathbb{Z}}$ its quadratic variation process being defined by

$$[M]_n = \sum_{r=-\infty}^n (M_r - M_{r-1})^2 \quad (2.9)$$

and analyse the $q/2$ -th moment of its increments $\Delta[M]_n := (M_n - M_{n-1})^2$. Note that for every $n \in \mathbb{Z} \cap (-\infty, N]$, one has that

$$\Delta M_n = \mathbb{E}[\bar{\Theta}_N | \mathcal{F}_n] - \mathbb{E}[\bar{\Theta}_N | \mathcal{F}_{n-1}] = \mathbb{E}\left[\sum_{r=n_0+1}^N \gamma_r g(\mathbf{X}(r)) | \mathcal{F}_n\right] - \mathbb{E}\left[\sum_{r=n_0+1}^N \gamma_r g(\mathbf{X}(r)) | \mathcal{F}_{n-1}\right]. \quad (2.10)$$

Now using that (X_n) is an i.i.d. sequence we conclude that, for almost all $\omega \in \Omega$,

$$\mathbb{E}\left[\sum_{r=n_0+1}^N \gamma_r g(\mathbf{X}(r)) | \mathcal{F}_n\right](\omega) = \mathbb{E}\left[\sum_{r=n_0+1}^N \gamma_r g(\mathbf{X}(r-n)) \middle| \forall k \in -\mathbb{N}_0 : X_k = X_{k+n}(\omega)\right] \quad (2.11)$$

and

$$\begin{aligned} \mathbb{E}\left[\sum_{r=n_0+1}^N \gamma_r g(\mathbf{X}(r)) | \mathcal{F}_{n-1}\right](\omega) &= \mathbb{E}\left[\sum_{r=n_0+1}^N \gamma_r g(\mathbf{X}(r-n)) \middle| \forall k \in -\mathbb{N} : X_k = X_{k+n}(\omega)\right] \\ &= \mathbb{E}\left[\sum_{r=n_0+1}^N \gamma_r g(\tilde{\mathbf{X}}(r-n)) \middle| \forall k \in -\mathbb{N}_0 : X_k = X_{k+n}(\omega)\right]. \end{aligned} \quad (2.12)$$

Consequently,

$$\begin{aligned} &\mathbb{E}\left[\sum_{r=n_0+1}^N \gamma_r g(\mathbf{X}(r)) | \mathcal{F}_n\right](\omega) - \mathbb{E}\left[\sum_{r=n_0+1}^N \gamma_r g(\mathbf{X}(r)) | \mathcal{F}_{n-1}\right](\omega) \\ &= \mathbb{E}\left[\underbrace{\sum_{r=n_0+1}^N \gamma_r (g(\mathbf{X}(r-n)) - g(\tilde{\mathbf{X}}(r-n)))}_{=: \Upsilon_n[N]} \middle| \forall k \in -\mathbb{N}_0 : X_k = X_{k+n}(\omega)\right]. \end{aligned} \quad (2.13)$$

We define the totaly influence of changing X_0 to $\tilde{X}_0$

$$\Upsilon_n[N] = \sum_{r=n_0+1}^N \gamma_r(g(\mathbf{X}(r-n)) - g(\tilde{\mathbf{X}}(r-n))) \quad (2.14)$$

Why we sum for r ? Becausr Adam has memory, and X_n will come to several future history:

$$\mathbf{X}_0, \mathbf{X}_1, \dots$$

Then by *Conditional Expection* and *Jensen*:

$$\|\Delta M_n\|_{L^p} \leq \|\Upsilon_n[N]\|_{L^p}$$

By stationarity, the distribution of the conditional expectation

$$\omega \mapsto \mathbb{E}[\Upsilon_n[N] | \forall k \in -\mathbb{N}_0 : X_k = X_{k+n}(\omega)] \quad (2.15)$$

is the same as the distribution of $\mathbb{E}[\Upsilon_n[N] | X_0, X_{-1}, \dots]$. Using this and Jensen's inequality we get that

$$\mathbb{E}[\Delta[M]_n^{p/2}] = \mathbb{E}[|M_n - M_{n-1}|^p] = \mathbb{E}[\mathbb{E}[\Upsilon_n[N] | X_0, X_{-1}, \dots]]^p \leq \mathbb{E}[|\Upsilon_n[N]|^p]. \quad (2.16)$$

It follows with the triangle inequality that

$$\mathbb{E}[(M_N^{p/2})^{2/p}] = \mathbb{E}\left[\left(\sum_{n=-\infty}^N \Delta[M]_n\right)^{p/2}\right]^{2/p} \leq \sum_{n=-\infty}^N \mathbb{E}[(\Delta[M]_n)^{p/2}]^{2/p} \leq \sum_{n=-\infty}^N \mathbb{E}[|\Upsilon_n[N]|^p]^{2/p}. \quad (2.17)$$

ϱ_{n-r} : How old X_n is in r -th update.

$|X_0 - \tilde{X}_0|$: the origin error of replacing the random sample.

Define the scale of the noise:

$$C := \mathbb{E}[|X_0 - \mathbb{E}[X_0]|^p]^{1/p}$$

As X_0 are independent to $\tilde{X}_0$, we have

$$X_0 - \tilde{X}_0 = (X_0 - \mathbb{E}(X_0)) - (\tilde{X}_0 - \mathbb{E}(X_0))$$

By Minkowski inequality

$$\|X_0 - \tilde{X}_0\|_{L^p} \leq \|X_0 - \mathbb{E}(X_0)\|_{L^p} + \|\tilde{X}_0 - \mathbb{E}(X_0)\|_{L^p} = 2C$$

Now for $n \in \mathbb{Z}$ with $n \leq N$ we get with Lemma 1.2.3 that

$$|\Upsilon_n[N]| \leq \sum_{r=n \vee (n_0+1)}^N \gamma_r |g(\mathbf{X}(r-n)) - g(\tilde{\mathbf{X}}(r-n))| \leq \sum_{r=n \vee (n_0+1)}^N \gamma_r \varrho_{n-r} |X_0 - \tilde{X}_0| \quad (2.18)$$

and again by the triangle inequality we have in terms of $C := \mathbb{E}[|X_0 - \mathbb{E}[X_0]|^p]^{1/p}$,

$$\begin{aligned} \mathbb{E}[|\Upsilon_n[N]|^p]^{1/p} &\leq \sum_{r=n \vee (n_0+1)}^N \gamma_r \varrho_{n-r} \mathbb{E}[|X_0 - \tilde{X}_0|^p]^{1/p} \\ &\leq \begin{cases} 2C \|\varrho\|_{\ell_1} \gamma_n, & \text{if } n \geq n_0 + 1, \\ 2C(1 - \sqrt{\beta})^{-1} \varrho_{n-(n_0+1)} \gamma_{n_0+1}, & \text{if } n \leq n_0. \end{cases} \end{aligned} \quad (2.19)$$

If $n \geq n_0 + 1$: Samples truly newly generated within the optimization interval.

$$\|\Upsilon_n[N]\|_{L^p} \leq 2C \varrho_n \sum_{r=n}^N \varrho_{n-r} \leq 2C \gamma_n \|\varrho\|_{\ell^1}, \quad \text{as } \{\gamma_n\} \text{ is decreasing}$$

Hence we obtain

$$2C \gamma_n \|\varrho\|_{\ell^1}, \quad \text{where } \|\varrho\|_{\ell^1} = \sum_{j \leq 0} \varrho_j.$$

If $n \leq n_0$ Old historical samples from before the start of optimization.

$$\begin{aligned} \|\Upsilon_n[N]\|_{L^p} &\leq 2C \gamma_{n_0+1} \sum_{r=n_0+1}^N \varrho_{n-r} \\ &\leq 2C \gamma_{n_0+1} \varrho_{n-(n_0+1)} \sum_{j=0}^{\infty} (\sqrt{\beta})^j \\ &\leq 2C(1 - \sqrt{\beta})^{-1} \varrho_{n-(n_0+1)} \gamma_{n_0+1}. \end{aligned}$$

with

$$\varrho_{k-1} \leq \sqrt{\beta} \varrho_k.$$

Together with (2.17) we conclude that

$$\begin{aligned} \mathbb{E}[|M|_N^{p/2}]^{2/p} &\leq \sum_{n=-\infty}^N \mathbb{E}[|\Upsilon_n[N]|^p]^{2/p} \\ &\leq 4C^2(1 - \sqrt{\beta})^{-2} \gamma_{n_0+1}^2 \sum_{n=-\infty}^{n_0} \varrho_{n-(n_0+1)}^2 + 4C^2 \|\varrho\|_{\ell_1}^2 \sum_{n=n_0+1}^N \gamma_n^2 \\ &\leq 4C^2 \left((1 - \sqrt{\beta})^{-2} \varrho_0 \|\varrho\|_{\ell_1} \gamma_{n_0+1}^2 + \|\varrho\|_{\ell_1}^2 \sum_{n=n_0+1}^N \gamma_n^2 \right). \end{aligned} \quad (2.20)$$

The statement follows with the Burkholder-Davis-Gundy inequality since $\bar{\Theta}_N = \mathbb{E}[\bar{\Theta}_N] + M_N$. $\square$

2.1.2 Proof of Lemma 2.1.2

Proof. Without loss of generality we can assume that $n_0 = 0$ since the general statement then follows by an index shift.

Suppose that $m \in \mathbb{N}$ and $2^{m-1} \leq N \leq 2^m - 1$. We call for every $k = 1, \dots, m$, all intervals of the form

$$[j, j + 2^{m-k}) \quad (j \in (2^{m-k+1}\mathbb{N}_0) \cap \{0, \dots, N\}) \quad (2.21)$$

and additionally the empty set an interval of the k -th layer. Note that by construction for every natural number $n \in \{0, \dots, N\}$ the interval $[0, n)$ can be uniquely written as pairwise disjoint union

$$[0, n) = [0, n_1) \cup [n_1, n_2) \cup \dots \cup [n_{m-1}, n_m) \quad (2.22)$$

with each $[n_{k-1}, n_k)$ being an interval of the k -th layer. Indeed, the representation is obtained for $n_k = 2^{m-k} \lfloor 2^{-(m-k)} n \rfloor$. We let $\bar{\Theta}_{k,\ell} := \bar{\Theta}_\ell - \bar{\Theta}_k - \mathbb{E}[\bar{\Theta}_\ell - \bar{\Theta}_k]$ for every $k, \ell \in \mathbb{N}_0$ with $0 \leq k \leq \ell$. Then for n as above we have

$$\bar{\Theta}_{0,n} = \sum_{k=1}^m \bar{\Theta}_{n_{k-1}, n_k}. \quad (2.23)$$

Consequently,

$$\max_{n \in \{0, \dots, N\}} |\bar{\Theta}_{0,n}| \leq \sum_{k=1}^m \max_{[n_{k-1}, n_k): \text{interval of } k\text{th layer}} |\bar{\Theta}_{n_{k-1}, n_k}| \quad (2.24)$$

and

$$\mathbb{E} \left[\max_{n \in \{0, \dots, N\}} |\bar{\Theta}_{0,n}|^p \right]^{1/p} \leq \sum_{k=1}^m \mathbb{E} \left[\max_{[n_{k-1}, n_k): \text{interval of } k\text{th layer}} |\bar{\Theta}_{n_{k-1}, n_k}|^p \right]^{1/p}. \quad (2.25)$$

Now note that by Lemma 2.1.1 one has in terms of $C := \mathbb{E}[|X_0 - \mathbb{E}[X_0]|^p]^{1/p}$ that for an arbitrary non-empty interval $[n_{k-1}, n_k)$ of the k -th layer

$$\mathbb{E}[|\bar{\Theta}_{n_{k-1}, n_k}|^p]^{2/p} \leq \kappa^2 \gamma_1^2 2^{m-k} C^2 \quad (2.26)$$

so that

$$\begin{aligned} \mathbb{E} \left[\max_{[n_{k-1}, n_k): \text{interval of } k\text{th layer}} |\bar{\Theta}_{n_{k-1}, n_k}|^p \right] &\leq \sum_{[n_{k-1}, n_k): \text{interval of } k\text{th layer}} \mathbb{E}[|\bar{\Theta}_{n_{k-1}, n_k}|^p] \\ &\leq \kappa^p \gamma_1^p C^p 2^{k-1} (2^{m-k})^{p/2}. \end{aligned} \quad (2.27)$$

Together with (2.25) we obtain that

$$\mathbb{E} \left[\max_{n \in \{0, \dots, N\}} |\bar{\Theta}_{0,n}|^p \right]^{1/p} \leq \kappa \gamma_1 C \sqrt{2^{m-1}} \sum_{k=1}^m 2^{-(\frac{1}{2} - \frac{1}{p})(k-1)} \leq (1 - 2^{-(\frac{1}{2} - \frac{1}{p})})^{-1} \kappa \gamma_1 C \sqrt{N}. \quad (2.28)$$

□

2.2 Approximation to Adam process

For this we fix damping parameters $\alpha, \beta, \epsilon \in [0, \infty)$ with $\alpha < \sqrt{\beta} < 1$ and an innovation (X, U) and denote by $(\theta_n)_{n \geq n_0}$ an Adam algorithm started at time $n_0 \in \mathbb{N}_0$ in a state $(\theta_{n_0}, m_{n_0}, v_{n_0})$. We assume that there exists a deterministic $\mathbf{x} = (x_k)_{k \in -\mathbb{N}_0} \in \ell^d_{\mathcal{Q}}$ such that for all $i \in \{1, \dots, d\}$ we have

$$m_{n_0} = (1 - \alpha) \sum_{k \in -\mathbb{N}_0} \alpha^{-k} x_k \quad \text{and} \quad v_{n_0}^{(i)} = (1 - \beta) \sum_{k \in -\mathbb{N}_0} \beta^{-k} (x_k^{(i)})^2. \quad (2.29)$$

We consider the $\mathbb{R}^d$ -valued process $(X_k)_{k \in \mathbb{Z}}$ given by

$$X_k = \begin{cases} x_{k-n_0}, & \text{if } k \leq n_0, \\ X(U_k, \theta_{k-1}), & \text{if } k > n_0 \end{cases} \quad (2.30)$$

First approximation:

Dropping out the correction term $(1 - \beta^n)$:

$$\Theta_n := \sum_{k=n_0+1}^n \gamma_k g(\mathbf{X}(k)). \quad (2.31)$$

Second approximation:

From n_0 , we cut the future terms with $(n_\ell)_{\ell \in \mathbb{N}_0}$. Thus we have *intervals* like:

$$\begin{aligned} &\{n_0 + 1, n_0 + 2, \dots, n_1\} \\ &\{n_1 + 1, n_1 + 2, \dots, n_2\} \\ &\vdots \end{aligned}$$

For every $\ell \in \mathbb{N}$ and $n \in \{n_{\ell-1} + 1, \dots, n_\ell\}$, (*understand n as stopping time, though not exactly*) the process $\tilde{\mathbf{X}}(n) = (\tilde{\mathbf{X}}(n)_k)_{k \in -\mathbb{N}_0}$ by

$$\tilde{\mathbf{X}}(n)_k = \begin{cases} X(U_{n+k}, \theta_{n_{\ell-1}}), & \text{if } n+k > n_{\ell-1}, \\ X_{n+k}, & \text{else.} \end{cases} \quad (2.32)$$

Comparing $\mathbf{X}(n)$ and $\tilde{\mathbf{X}}(n)$ we see that the terms outside the $(n_{\ell-1}, n_\ell]$ -window agree and inside the window the θ -parameter is fixed as $\theta_{n_{\ell-1}}$. In analogy to before we consider the process $(\tilde{\Theta}_n)_{n \geq n_0}$ given by

$$\tilde{\Theta}_n := \sum_{k=n_0+1}^n \gamma_k g(\tilde{\mathbf{X}}(k)). \quad (2.33)$$

Third approximation

Next, let $(U_{\ell,n})_{\ell \in \mathbb{N}, n \in \mathbb{Z}}$ be a family of independent copies of U that is also independent of $(U_n)_{n \in \mathbb{N}}$. For every $\ell \in \mathbb{N}$ and $n \in \{n_{\ell-1} + 1, \dots, n_\ell\}$, we consider $\tilde{\tilde{\mathbf{X}}}(n) = (\tilde{\tilde{\mathbf{X}}}(n)_k)_{k \in -\mathbb{N}_0}$

given by

$$\tilde{\tilde{\mathbf{X}}}(n)_k = \begin{cases} X(U_{n+k}, \theta_{n_{\ell-1}}) = \tilde{\mathbf{X}}(n)_k, & \text{if } n+k > n_{\ell-1}, \\ X(U_{\ell, n+k}, \theta_{n_{\ell-1}}), & \text{else.} \end{cases} \quad (2.34)$$

In analogy to before we define a process $(\tilde{\tilde{\Theta}}_n)_{n \geq n_0}$ via

$$\tilde{\tilde{\Theta}}_n = \sum_{k=n_0+1}^n \gamma_k g(\tilde{\tilde{\mathbf{X}}}(k)). \quad (2.35)$$

Understand the third approximation as re-write the whole history with the location $\theta_{n_{\ell-1}}$.

Then we come to the result of approximation:

Proposition 2.2.1. *Let*

- (i) $\alpha, \beta \in [0, 1)$ with $\alpha < \sqrt{\beta}$ and $\epsilon \in (0, \infty)$ (the damping parameters),
- (ii) $n_0 \in \mathbb{N}_0$, $\theta_{n_0}, m_{n_0} \in \mathbb{R}^d$, $v_{n_0} \in [0, \infty)^d$ (the initialisation),
- (iii) a decreasing $(0, \infty)$ -valued sequence $(\gamma_n)_{n \in \mathbb{N}}$ (sequence of step-sizes),
- (iv) (X, U) an innovation,
- (v) $C, \tilde{C}, \tilde{L} \in [0, \infty)$, $p \in [2, \infty)$ and $V \subset \mathbb{R}^d$ be a measurable set,
- (vi) $(n_\ell)_{\ell \in \mathbb{N}_0}$ be a strictly increasing $\mathbb{N}_0$ -valued sequence.

Let (X, U) be a p -regular innovation with parameter $(C, \tilde{C}, \tilde{L})$ on V and $(\theta_n)_{n \in \mathbb{N}_0 \cap [n_0, \infty)}$ the Adam algorithm with damping parameter $(\alpha, \beta, \epsilon)$ and step-sizes (γ_n) started at time n_0 in $(\theta_{n_0}, m_{n_0}, v_{n_0})$. Moreover, let

$$\mathfrak{N} = \inf\{n \geq n_0 : \theta(n) \notin V\} \quad (2.36)$$

and let $(t_n)_{n \in \mathbb{N}}$ as in (1.2). One has the following for the approximations $(\Theta_n)_{n \geq n_0}$, $(\tilde{\Theta}_n)_{n \geq n_0}$ and $(\tilde{\tilde{\Theta}}_n)_{n \geq n_0}$:

(I) for every $\mathbf{n} \in \{n_0, n_0 + 1, \dots\}$,

$$\mathbb{E}\left[\left(\sum_{k=\mathbf{n}+1}^{\infty} \mathbb{1}_{\{\mathfrak{N} \geq k\}} |\Delta \theta_k - \Delta \Theta_k|\right)^p\right]^{1/p} \leq (\kappa_1 \beta^{\frac{1}{2}(\mathbf{n}+1)-n_0} \|\mathbf{x}\|_{\ell_{\mathbf{g}}^d} + \kappa_2 C) \gamma_{\mathbf{n}+1} \beta^{\mathbf{n}+1} \quad (2.37)$$

(II) for every $\ell \in \mathbb{N}$ and $\mathbf{n}, n \in \mathbb{N}_0$ with $n_{\ell-1} \leq \mathbf{n} \leq n \leq n_\ell$,

$$\mathbb{E}\left[\left(\sum_{k=\mathbf{n}+1}^n \mathbb{1}_{\{\mathfrak{N} \geq k\}} |\Delta \Theta_k - \Delta \tilde{\Theta}_k|\right)^p\right]^{1/p} \leq \kappa_3 \tilde{L}(t_{n_\ell} - t_{n_{\ell-1}})(t_n - t_{\mathbf{n}}) \quad (2.38)$$

(III) for every $\ell \in \mathbb{N}$,

$$\mathbb{E}\left[\mathbb{1}_{\{\mathfrak{N} > n_{\ell-1}\}} \left(\sum_{k=n_{\ell-1}+1}^{n_\ell} |\Delta \tilde{\Theta}_k - \Delta \tilde{\tilde{\Theta}}_k|\right)^p\right]^{1/p} \leq \gamma_{n_{\ell-1}+1} (\kappa_4 C + \kappa_5 \beta^{(n_{\ell-1}-n_0)/2} \|\mathbf{x}\|_{\ell_{\mathbf{g}}^d}), \quad (2.39)$$

(IV.a) for every $\ell \in \mathbb{N}$,

$$\mathbb{E}[\mathbb{1}_{\{\mathfrak{N} > n_{\ell-1}\}} |\tilde{\Theta}_{n_{\ell}} - \tilde{\Theta}_{n_{\ell-1}} - (t_{n_{\ell}} - t_{n_{\ell-1}})f(\theta_{n_{\ell-1}})|^p | \mathcal{F}_{n_{\ell-1}}]^{1/p} \leq \kappa_6 \tilde{C} \left(\sum_{k=n_{\ell-1}+1}^{n_{\ell}} \gamma_k^2 \right)^{1/2}, \quad (2.40)$$

(IV.b) if $p > 2$, then for every $\ell \in \mathbb{N}$,

$$\begin{aligned} \mathbb{E}[\mathbb{1}_{\{\mathfrak{N} > n_{\ell-1}\}} \max_{n=n_{\ell-1}, \dots, n_{\ell}} |\tilde{\Theta}_n - \tilde{\Theta}_{n_{\ell-1}} - (t_n - t_{n_{\ell-1}})f(\theta_{n_{\ell-1}})|^p | \mathcal{F}_{n_{\ell-1}}]^{1/p} \\ \leq (1 - 2^{-(\frac{1}{2} - \frac{1}{p})})^{-1} \kappa_6 \tilde{C} \gamma_{n_{\ell-1}+1} \sqrt{n_{\ell} - n_{\ell-1}}, \end{aligned} \quad (2.41)$$

where

$$\begin{aligned} \kappa_1 = \frac{2}{1 - \beta^{3/2}}, \quad \kappa_2 = \frac{2}{1 - \beta^{3/2}} \left(\frac{\varrho_0}{1 - \beta} + \|\varrho\|_{\ell_1} \right), \quad \kappa_3 = \frac{1 - \alpha}{\sqrt{1 - \beta} \sqrt{1 - \alpha^2/\beta}} d \|\varrho\|_{\ell_1}, \\ \kappa_4 = 2(1 - \sqrt{\beta})^{-1} \|\varrho\|_{\ell_1}, \quad \kappa_5 = (1 - \sqrt{\beta})^{-1}, \quad \kappa_6 = 2C_p \left((1 - \sqrt{\beta})^{-1} \sqrt{\varrho_0 \|\varrho\|_{\ell_1}} + \|\varrho\|_{\ell_1} \right), \end{aligned} \quad (2.42)$$

(ϱ_k) is as in (1.4) and C_p is the constant in the Burkholder-Davis-Gundy inequality when applied for the p th moment.

Proof. 1) We prove the first inequality. Note that

$$\begin{aligned} |\sigma_k^{(i)} m_k^{(i)} - g(\mathbf{X}^{(i)}(k))| &\leq \left(1 - \frac{1}{\sqrt{1/(1 - \beta^k)}} \right) |g(\mathbf{X}^{(i)}(k))| \leq 2\beta^k \|\mathbf{X}^{(i)}(k)\|_{\ell_{\varrho}} \\ &\leq 2\beta^k \left(\sqrt{\beta^{k-n_0}} \|\mathbf{x}^{(i)}\|_{\ell_{\varrho}} + \sum_{r=n_0+1}^k \varrho_{r-k} |X_r^{(i)}| \right). \end{aligned} \quad (2.43)$$

This entails that

$$\mathbb{E}[\mathbb{1}_{\{\mathfrak{N} \geq k\}} |\Delta\theta_k - \Delta\Theta_k|^p]^{1/p} \leq 2\beta^k \left(\sqrt{\beta^{k-n_0}} \|\mathbf{x}\|_{\ell_{\varrho}^d} + C \sum_{r=n_0+1}^k \varrho_{r-k} \right). \quad (2.44)$$

Consequently,

$$\begin{aligned} \mathbb{E} \left[\left(\sum_{k=\mathfrak{n}+1}^{\infty} \mathbb{1}_{\{\mathfrak{N} \geq k\}} \gamma_k |\Delta\theta_k - \Delta\Theta_k| \right)^p \right]^{1/p} &\leq 2 \sum_{k=\mathfrak{n}+1}^{\infty} \gamma_k \beta^k \left(\sqrt{\beta^{k-n_0}} \|\mathbf{x}\|_{\ell_{\varrho}^d} + C \sum_{r=n_0+1}^k \varrho_{r-k} \right) \\ &\leq 2(1 - \beta^{3/2})^{-1} \gamma_{\mathfrak{n}+1} \beta^{\frac{3}{2}(\mathfrak{n}+1)-n_0} \|\mathbf{x}\|_{\ell_{\varrho}^d} + 2C \gamma_{\mathfrak{n}+1} \sum_{r=n_0+1}^{\infty} \sum_{k=r \vee (\mathfrak{n}+1)}^n \beta^k \varrho_{r-k} \end{aligned} \quad (2.45)$$

In the case where $r \leq \mathfrak{n}$, one has that

$$\sum_{k=r \vee (\mathfrak{n}+1)}^n \beta^k \varrho_{r-k} \leq (1 - \beta^{3/2})^{-1} \varrho_{r-(\mathfrak{n}+1)} \beta^{\mathfrak{n}+1} \quad (2.46)$$

and in the case where $r > \mathfrak{n}$,

$$\sum_{k=r \vee (\mathfrak{n}+1)}^n \beta^k \varrho_{r-k} \leq (1 - \beta^{3/2})^{-1} \varrho_0 \beta^r. \quad (2.47)$$

Hence,

$$\begin{aligned}
& \mathbb{E} \left[\left(\sum_{k=\mathbf{n}+1}^{\infty} \mathbb{1}_{\{\mathfrak{N} \geq k\}} \gamma_k |\Delta \theta_k - \Delta \Theta_k| \right)^p \right]^{1/p} \\
& \leq 2(1 - \beta^{3/2})^{-1} \gamma_{\mathbf{n}+1} \left(\beta^{\frac{3}{2}(\mathbf{n}+1)-n_0} \|\mathbf{X}\|_{\ell_{\varrho}^d} + \beta^{\mathbf{n}+1} C \sum_{r=n_0+1}^{\mathbf{n}} \varrho_{r-(\mathbf{n}+1)} + C \sum_{r=\mathbf{n}+1}^n \varrho_0 \beta^r \right) \\
& \leq 2(1 - \beta^{3/2})^{-1} \left(\beta^{\frac{1}{2}(\mathbf{n}+1)-n_0} \|\mathbf{X}\|_{\ell_{\varrho}^d} + (\|\varrho\|_{\ell_1} + (1 - \beta)^{-1} \varrho_0) C \right) \gamma_{\mathbf{n}+1} \beta^{\mathbf{n}+1}.
\end{aligned} \tag{2.48}$$

2.) Fix $\ell \in \mathbb{N}$. Since for every $k \geq n_0$

$$\sigma_k^{(i)} m_k^{(i)} \leq \frac{1 - \alpha}{\sqrt{1 - \beta}} \frac{1}{\sqrt{1 - \alpha^2/\beta}}, \tag{2.49}$$

one has for every $n = n_{\ell-1} + 1, \dots, n_{\ell}$

$$|\theta_n^{(i)} - \theta_{n_{\ell-1}}^{(i)}| = \left| \sum_{k=n_{\ell-1}+1}^n \gamma_k \sigma_k^{(i)} m_k^{(i)} \right| \leq \frac{1 - \alpha}{\sqrt{1 - \beta}} \frac{1}{\sqrt{1 - \alpha^2/\beta}} (t_n - t_{n_{\ell-1}}). \tag{2.50}$$

This implies that

$$\mathbb{E} [\mathbb{1}_{\{\mathfrak{N} \geq n\}} |X(U_n, \theta_{n-1}) - X(U_n, \theta_{n_{\ell-1}})|^p | \mathcal{F}_{n-1}]^{1/p} \leq \underbrace{\frac{1 - \alpha}{\sqrt{1 - \beta}} \frac{1}{\sqrt{1 - \alpha^2/\beta}}}_{=\kappa_3/\|\varrho\|_{\ell_1}} d\tilde{L}(t_{n_{\ell}} - t_{n_{\ell-1}}). \tag{2.51}$$

Now note that for $k = n_{\ell-1} + 1, \dots, n_{\ell}$

$$|g(\mathbf{X}(k)) - g(\tilde{\mathbf{X}}(k))| \leq \sum_{r=n_{\ell-1}+1}^k \rho_{r-k} |X(U_r, \theta_{r-1}) - X(U_r, \theta_{n_{\ell-1}})|. \tag{2.52}$$

This implies that for $\mathbf{n}, n \in \mathbb{N}_0$ with $n_{\ell-1} \leq \mathbf{n} \leq n \leq n_{\ell}$

$$\begin{aligned}
& \mathbb{E} \left[\left(\sum_{k=\mathbf{n}+1}^n \mathbb{1}_{\{\mathfrak{N} > k\}} \gamma_k |g(\mathbf{X}(k)) - g(\tilde{\mathbf{X}}(k))| \right)^p \right]^{1/p} \\
& \leq \sum_{k=\mathbf{n}+1}^n \gamma_k \mathbb{E} \left[\left(\mathbb{1}_{\{\mathfrak{N} > k\}} |g(\mathbf{X}(k)) - g(\tilde{\mathbf{X}}(k))| \right)^p \right]^{1/p} \\
& \leq \sum_{k=\mathbf{n}+1}^n \gamma_k \mathbb{E} \left[\left(\mathbb{1}_{\{\mathfrak{N} > k\}} \sum_{r=n_{\ell-1}+1}^k \rho_{r-k} |X(U_r, \theta_{r-1}) - X(U_r, \theta_{n_{\ell-1}})| \right)^p \right]^{1/p} \\
& \leq \sum_{k=\mathbf{n}+1}^n \gamma_k \sum_{r=n_{\ell-1}+1}^k \varrho_{r-k} \mathbb{E} \left[\left(\mathbb{1}_{\{\mathfrak{N} > r\}} |X(U_r, \theta_{r-1}) - X(U_r, \theta_{n_{\ell-1}})| \right)^p \right]^{1/p} \\
& \leq \kappa_3 \tilde{L}(t_{n_{\ell}} - t_{n_{\ell-1}}) \sum_{k=\mathbf{n}+1}^n \gamma_k.
\end{aligned} \tag{2.53}$$

3.) Note that

$$|\Delta \tilde{\Theta}_{\mathbf{n},n} - \Delta \tilde{\Theta}_{\mathbf{n},n}| = |\gamma_n (g(\tilde{\mathbf{X}}(n)) - g(\tilde{\mathbf{X}}(n)))| \leq \gamma_n \|\tilde{\mathbf{X}}(n) - \tilde{\mathbf{X}}(n)\|_{\ell_{\varrho}^d} \tag{2.54}$$

and for $n = n_{\ell-1} + 1, \dots, n_\ell$,

$$\begin{aligned} \|\tilde{\tilde{\mathbf{X}}}(n) - \tilde{\mathbf{X}}(n)\|_{\ell_\varrho^d} &= \sum_{k=-\infty}^{n_{\ell-1}} \varrho_{k-n} |X_k - X(U_{\ell,k}, \theta_{n_{\ell-1}})| \\ &\leq \sum_{k=-\infty}^{n_{\ell-1}} \varrho_{k-n} |X_k| + \sum_{k=-\infty}^{n_{\ell-1}} \varrho_{k-n} |X(U_{\ell,k}, \theta_{n_{\ell-1}})|. \end{aligned} \quad (2.55)$$

Hence,

$$\begin{aligned} \mathbb{E}[\mathbb{1}_{\{\mathfrak{N} > n_{\ell-1}\}} \|\tilde{\tilde{\mathbf{X}}}(n) - \tilde{\mathbf{X}}(n)\|_{\ell_\varrho^d}^p]^{1/p} &\leq \|\mathbf{x}\|_{\ell_\varrho^d} \sqrt{\beta}^{n-n_0} + 2C \sum_{k=-\infty}^{n_{\ell-1}} \varrho_{k-n} \\ &\leq \|\mathbf{x}\|_{\ell_\varrho^d} \sqrt{\beta}^{n-n_0} + 2(1 - \sqrt{\beta})^{-1} C \varrho_{n_{\ell-1}-n} \end{aligned} \quad (2.56)$$

and

$$\begin{aligned} \mathbb{E}[\mathbb{1}_{\{\mathfrak{N} > n_{\ell-1}\}} \sum_{n=n_{\ell-1}+1}^{n_\ell} |\Delta \tilde{\tilde{\Theta}}_n - \Delta \tilde{\Theta}_n|^p]^{1/p} \\ \leq \gamma_{n_{\ell-1}+1} \sum_{n=n_{\ell-1}+1}^{n_\ell} (\|\mathbf{x}\|_{\ell_\varrho^d} \sqrt{\beta}^{n-n_0} + 2(1 - \sqrt{\beta})^{-1} C \varrho_{n_{\ell-1}-n}) \\ \leq \frac{1}{1 - \sqrt{\beta}} \gamma_{n_{\ell-1}+1} \sqrt{\beta}^{n_{\ell-1}-n_0} \|\mathbf{x}\|_{\ell_\varrho^d} \\ + \frac{2}{1 - \sqrt{\beta}} \gamma_{n_{\ell-1}+1} \|\varrho\|_{\ell_1} C. \end{aligned} \quad (2.57)$$

4.) The inequalities (IV.a) and (IV.b) are direct consequences of Lemmas 2.1.1 and 2.1.2 when conditioning on $\mathcal{F}_{n_{\ell-1}}$. $\square$

When choosing in estimate (II), $\mathbf{n} = n_{\ell-1}$ and $n = n_\ell$ we obtain an estimate covering the same range of indices as the estimate (III). In order to obtain good results we will choose the sequence $(n_\ell)_{\ell \in \mathbb{N}_0}$ in such a way that both error terms are of the same order. This motivates the following definition.

Definition 2.2.2.

Let $n_0 \in \mathbb{N}_0$ and $\rho \in [\sqrt{\gamma_{n_0+1}}, \infty)$. We call the $\mathbb{N}_0$ -valued sequence $(n_\ell)_{\ell \in \mathbb{N}_0}$ a ρ -partition w.r.t. (γ_n) if for every $\ell \in \mathbb{N}$, n_ℓ is the largest integer with

$$t_{n_\ell} - t_{n_{\ell-1}} \leq \rho \sqrt{\gamma_{n_{\ell-1}+1}}, \quad (2.58)$$

where (t_n) is as in (1.2). We call n_0 the starting value.

Remark 2.2.3. In the definition we assume that $\rho \geq \sqrt{\gamma_{n_0+1}}$. This guarantees that for every $\ell \in \mathbb{N}$, one has $\rho \sqrt{\gamma_{n_{\ell-1}+1}} \geq \gamma_{n_{\ell-1}+1}$ by monotonicity of (γ_n) so that $n_\ell > n_{\ell-1}$. This implies that (n_ℓ) is strictly increasing and that $\lim_{n \rightarrow \infty} n_\ell = \infty$.

Lemma 2.2.4. Let $(\gamma_n)_{n \in \mathbb{N}}$ be a $(0, \infty)$ -valued decreasing non-summable sequence and let $(n_\ell)_{\ell \in \mathbb{N}_0}$ be a ρ -partition for (γ_n) .

(i) For $\ell \in \mathbb{N}$ with $\gamma_{n_{\ell-1}+1} < \rho^2$, one has that

$$\sqrt{\gamma_{n_{\ell-1}+1}} \leq \frac{1}{\rho - \sqrt{\gamma_{n_{\ell-1}+1}}} (t_{n_{\ell}} - t_{n_{\ell-1}}). \quad (2.59)$$

(ii) Assume that for all $n = n_0 + 1, n_0 + 2, \dots$

$$\frac{\gamma_n - \gamma_{n+1}}{\gamma_n^2} \leq \zeta \quad (2.60)$$

and that $\rho\zeta\sqrt{\gamma_{n_0+1}} < 1$. Set $K = (1 - \zeta\rho\sqrt{\gamma_{n_0+1}})^{-1}$. One has for every $\ell \in \mathbb{N}$ that

$$\frac{\gamma_{n_{\ell-1}+1}}{\gamma_{n_{\ell}+1}} \leq K \quad \text{and} \quad \frac{\gamma_{n_{\ell-1}+1}}{\gamma_{n_{\ell}+1}} \leq 1 + \zeta K (t_{n_{\ell}} - t_{n_{\ell-1}}). \quad (2.61)$$

Proof. (i): This is an immediate consequence of

$$t_{n_{\ell}} - t_{n_{\ell-1}} \geq \rho\sqrt{\gamma_{n_{\ell-1}+1}} - \gamma_{n_{\ell-1}+1} = (\rho - \sqrt{\gamma_{n_{\ell-1}+1}})\sqrt{\gamma_{n_{\ell-1}+1}}. \quad (2.62)$$

(ii): Denote by $f : [0, \infty) \rightarrow (0, \infty)$ the function satisfying

$$f(t_n) = \gamma_{n+1} \quad (2.63)$$

for every $n \in \mathbb{N}_0$ and that is piecewise linear in between the points of $\{t_0, t_1, \dots\}$. First note that on each interval (t_{n-1}, t_n)

$$-f'(s) = \frac{\gamma_n - \gamma_{n+1}}{\gamma_n} \leq \zeta\gamma_n \quad (2.64)$$

so that, for $\ell \in \mathbb{N}$,

$$\gamma_{n_{\ell}+1} \geq \gamma_{n_{\ell-1}+1} - \zeta\gamma_{n_{\ell-1}+1}\rho\sqrt{\gamma_{n_{\ell-1}+1}} \geq (1 - \zeta\rho\sqrt{\gamma_{n_0+1}})\gamma_{n_{\ell-1}+1} \quad (2.65)$$

and

$$\frac{\gamma_{n_{\ell-1}+1}}{\gamma_{n_{\ell}+1}} \leq K. \quad (2.66)$$

One has

$$\gamma_{n_{\ell-1}+1} - \gamma_{n_{\ell}+1} = f(t_{n_{\ell-1}}) - f(t_{n_{\ell}}) \leq \zeta\gamma_{n_{\ell-1}+1}(t_{n_{\ell}} - t_{n_{\ell-1}}) \quad (2.67)$$

and

$$\frac{\gamma_{n_{\ell-1}+1} - \gamma_{n_{\ell}+1}}{\gamma_{n_{\ell}+1}} \leq \zeta K (t_{n_{\ell}} - t_{n_{\ell-1}}). \quad (2.68)$$

□

We provide another technical estimate.

Lemma 2.2.5. *Let $\|\cdot\|$ be a norm induced by a scalar product $\langle\langle\cdot,\cdot\rangle\rangle$ on $\mathbb{R}^d$ and let $p \geq 2$. One has for every $x, y \in \mathbb{R}^d$ that*

$$\|x + y\|^p \leq \|x\|^p + p\|x\|^{p-2}\langle\langle x, y \rangle\rangle + \frac{1}{2}p(p-1)(\|x\| \vee \|x + y\|)^{p-2}\|y\|^2. \quad (2.69)$$

Proof. Let $x, y \in \mathbb{R}^d$. Note that the function $h(z) = \|z\|^p$ ($z \in \mathbb{R}^d$) is twice continuously differentiable with differentials

$$Dh(z)(a) = p\|z\|^{p-2}\langle\langle z, a \rangle\rangle \quad \text{and} \quad D^2h(z)(a, b) = p(p-2)\|z\|^{p-4}\langle\langle z, a \rangle\rangle\langle\langle z, b \rangle\rangle + p\|z\|^{p-2}\langle\langle a, b \rangle\rangle. \quad (2.70)$$

By Taylor's theorem, we thus have

$$|x + y|^p = |x|^p + p|x|^{p-2}\langle x, y \rangle + \int_0^1 D^2h(x + ry)(y, y)(1 - r) \, dr. \quad (2.71)$$

Note that for every $r \in [0, 1]$, one has $|x + ry| \leq |x| \vee |x + y|$ and

$$\begin{aligned} \|D^2h(x + ry)(y, y)\| &\leq p(p-2)\|x + ry\|^{p-4}|\langle\langle x + ry, y \rangle\rangle\langle\langle x + ry, y \rangle\rangle| + p\|x + ry\|^{p-2}\langle\langle y, y \rangle\rangle \\ &\leq p(p-1)(\|x\| \vee \|x + y\|)^{p-2}\|y\|^2. \end{aligned} \quad (2.72)$$

The statement follows since $\int_0^1 (1 - r) \, dr = \frac{1}{2}$. □

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